

Polynomial operations and basic factoring

While TSM 099 studies Topic 17. Review: Topic 9, Topic 10, Topic 11, Topic 12, Topic 14, Topic 15.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: like terms and distribution

1. Combine like terms.

$$4x^2 - x$$

2. Combine like terms.

$$9x^2 + 8x + 1$$

3. Combine like terms.

$$-x^2 - 5x - 5$$

4. Distribute and simplify.

$$10x^3 + 20x^2 - 10x$$

5. Distribute and simplify.

$$5x^3 + 20x^2 - 25x$$

6. Distribute and simplify.

$$3x^3 + 9x^2 + 15x$$

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Core: polynomial multiplication

7. Multiply and combine like terms.

$$2x^2 + 6x - 8$$

8. Multiply and combine like terms.

$$x^2 - 16$$

9. Multiply and combine like terms.

$$x^2 + 2x - 3$$

10. Multiply and combine like terms.

$$x^2 + 7x + 12$$

11. Multiply and combine like terms.

$$x^4 + 3x^3 + 7x^2 + 7x + 6$$

12. Multiply and combine like terms.

$$x^4 - 3x^2 + 4x - 3$$

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Core: introductory factoring

13. Factor completely; check by multiplying.

$$(x + 3)(x + 4)$$

14. Factor completely; check by multiplying.

$$(x - 2)(x - 6)$$

15. Factor completely; check by multiplying.

$$(x + 5)(x - 3)$$

16. Factor completely; check by multiplying.

$$(x - 8)(x + 2)$$

17. Factor completely; check by multiplying.

$$6x(x + 2)$$

18. Factor completely; check by multiplying.

$$(x - 4)(x + 4)$$

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Extension: mixed polynomial practice

19. Multiply and simplify.

$$x^4 + 3x^3 + x^2 - 3x - 2$$

20. Multiply and simplify.

$$2x^4 - x^3 + 5x - 6$$

21. Multiply and simplify.

$$x^3 + 2x^2 - 4x - 8$$

22. Factor completely and verify.

$$(x + 4)(x - 7)$$

23. Factor completely and verify.

$$(x - 5)(x - 5)$$

24. Explain the error: only the first and last terms were multiplied.

$$(x + 2)(x + 3) = x^2 + 5x + 6$$

Every term in one factor must multiply every term in the other. The missing middle products are $3x$ and $2x$.